Harsh Meel

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EDUCATION

Cornell University, Sibley School of Mechanical and Aerospace Engineering, Ithaca, USA Aug 2024 – May 2026

Master of Science (Research) in Mechanical Engineering | Concentration: Dynamics and Control, Atmospheric Science

Indian Institute of Technology Bombay (IIT Bombay), Mumbai, India 2019 - 2023

Bachelor of Technology in Civil Engineering with Minor in Mechanical Engineering

KEY EXPERIENCE

Regional Climate-Informed Usage of Geoengineering, Climate Engineering Lab | MS Thesis [Oct'24- Current]

Principal Advisor: Prof. Douglas MacMartin, MAE, Cornell University; Minor Advisor: Prof. Daniele Visioni, CALS, Cornell University

- Researching controlled Stratospheric Aerosol Injection (SAI) as a potential climate change intervention strategy
- Developing metrics to evaluate SAI's impacts on regional climate and ecosystems, with focus on Indian Monsoon
- Analyzing simulation results to design strategies that stabilize climate while minimizing disagreeable effects of SAI
- Using robust decision-making for multi-scenario analysis, formulating ways to detect and respond to uncertainties
- Mapping control strategies and model outputs to physical climate dynamics for process attribution and validation
- Aiming to publish in a peer-reviewed journal to inform researchers and policymakers on SAI feasibility and risks

Student Electric Race Car Team, Formula Student | Accumulator (Battery) Engineer [Rep] [Nov'19-Jun'22]

IIT Bombay Racing - 3-tier cross-functional team of 70+ undergrad students building an electric race car for Formula Student. The team won Design Event'21 and Overall Championship'21 Concept Class at Formula Student UK, pioneer among Asian teams

- Responsible for design, analysis, manufacturing and testing of 400V battery made of composite material and encompassing lithium-polymer cell modules having capacity of 7.8 kWh and worth INR 1Million+
- Developed a new method to calculate energy required by vehicle using driver performance data and transient simulation of car, theoretically cutting battery weight by ~33%, reducing car's weight and increasing performance
- Designed a new battery configuration of parallel assembly of cells to reduce battery size and heat generation
- Initiated technical alliance with cell recyclers, reducing 640 kg eq. of annual CO_2 emissions from battery lifecycle
- Trained and managed 3 junior design engineers, acquainting them with design and subsystem management
- Led 4-member team to prepare and conduct recruitment tests for 140+ new aspirants for the engineering team

Operations Research for Electric Vehicle (EV) Charging Sector in India | Research Assistant [Nov'23 –June'24]

Guided By: Prof. Ashutosh Mahajan, Industrial Engineering and Operations Research, IIT Bombay, India

- Analyzed application of Minotaur toolkit for solving Mixed-Integer Nonlinear optimization problems in EV sector
- Surveyed state of EV charging in India by interacting with small and large businesses to identify key obstructions
- Summarized possible opportunities and limitations of operations research tools in aiding the EV charging penetration in India based on optimization of cost of building infrastructure, usage, and charger location/spread
- Tested the Minotaur Solver using a case study of Investment Portfolio Selection using Markowitz optimization

RELEVANT COURSES UNDERTAKEN

**Ongoing*

Graduate Level (Cornell)	Feedback Control Systems (MAE 5780) *, Environmental Statistics and Learning (BEE 6310) Climate Dynamics (EAS 5051), Aerosol and Climate Physics (EAS 6920), Energy Seminar
Undergrad (IIT Bombay)	Intro.to Renewable Energy Technologies, Environmental Studies: Science and Engineering, Fundamentals of Urban Science, Manufacturing Processes, Kinematics and Dynamics of Machines, Transportation Engineering, Thermodynamics, Mechanics (Fluid and Solid)

TECHNICAL/SOFTWARE SKILLS

Computational Skills	Python, MATLAB, Optimization (Pyomo, AMPL), SolidWorks, Ansys (Structural)
Prototyping/Machining	Additive Manufacturing, Electro-Mechanical Assembly, Metal and Composite Fabrication

RELEVANT PROJECTS

Price Reliability Analysis of Renewable Grid Transition Using Data-Driven Models [Report] [Nov'24 – Dec'24]

Guided by: Prof. Scott Steinschneider, Biological and Environmental Engineering, Cornell University

- Analyzed drivers of electricity price spikes in NY State by transition to renewable grid using load and fuel mix data
- Investigated service reliability under high renewable penetration; modeled price spikes using predictive analytics
- Applied regression, decision trees, and ensemble methods to find fit between spike events and the renewable grid
- Addressed imbalanced datasets and collinearity through feature engineering, correlation analysis, and VIF checks
- Derived insights on the result, identified data imbalance, multicollinearity, and external factors for better analysis

Optimizing Electric Vehicle (EV) Charging Infrastructure | BTech Technical Project [Report] [Jan'23 – April'23]

Guided by: Prof. Tom V. Mathew, Department of Civil Engineering, IIT Bombay

- Conducted comprehensive literature review, applied optimization theory for EV charger placement on highways
- Implemented a corridor travel model, with variable EV battery capacity, charger power, and time lost in queues

Residential Vehicle Charging Entrepreneurial Venture | Course Project [Report] [Aug'22 – Dec'22]

Guided by: Prof. Anuradha Narasimhan, Desai Sethi School of Entrepreneurship, IIT Bombay

- Formulated a home charging solution for electric vehicles, as a case study of technology's lab-to-market rollout
- Surveyed customers for usage behavior and installation challenges; outlined business model and retail strategy

Overtake Assist System | Institute Technical Summer Project [Report] [May'20 – Aug'20]

Tinkerers' Lab, IIT Bombay; Awarded Special Mention (top 7) out of 64 projects with over 200+ participants from the institute

- Built a vehicle-mountable driver assistance system, scrutinizing three-car overtake maneuvers on two-laned roads, assisting the driver to avoid head-on collisions in difficult weather and poor lighting conditions

PROFESSIONAL EXPERIENCE

ideaForge Technology | R&D Intern [May'22 – June'22]

Global leader in UAV technology | Developing drone solutions for Defense & Homeland Security, Surveying, and Enterprises

- Designed plan of modular, disassemblable wind tunnel capable of variable controllable flow in a confined space
- Developed testing setup to collect data and analyze flow of tube axial fans and relation of fan-flow parameters

AWARDS

- Awarded Swami Vivekanand Scholarship by Govt. of Rajasthan, India, to pursue graduate studies at Cornell [2025]